Therefore $Var(X) = E(X^2) - \left(E(X)\right)^2$

$$= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

Hence the mean and variance are respectively $\frac{1}{\lambda}$ and $\frac{1}{\lambda^2}$.

EXERCISE 11.4

1. For the random variable X with the given probability mass function as below, find the mean and variance.

(i) $f(x) = \begin{cases} \frac{1}{10} & x = 2, 5 \\ \frac{1}{5} & x = 0, 1, 3, 4 \end{cases}$

(ii) $f(x) = \begin{cases} \frac{4-x}{6} & x = 1, 2, 3 \end{cases}$

(iii) $f(x) = \begin{cases} 2(x-1) & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

(iv) $f(x) = \begin{cases} \frac{1}{2} e^{-\frac{x}{2}} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$

2. Two balls are drawn in succession without replacement from an urn containing four red balls and three black balls. Let X be the possible outcomes drawing red balls. Find the probability mass function and mean for X.
3. If μ and σ^2 are the mean and variance of the discrete random variable X, and $E(X+3) = 10$ and $E(X+3)^2 = 116$, find μ and σ^2.
4. Four fair coins are tossed once. Find the probability mass function, mean and variance for number of heads occurred.
5. A commuter train arrives punctually at a station every half hour. Each morning, a student leaves his house to the train station. Let X denote the amount of time, in minutes, that the student waits for the train from the time he reaches the train station. It is known that the pdf of X is

$$f(x) = \begin{cases} \frac{1}{30} & 0 < x < 30 \\ 0 & \text{elsewhere} \end{cases}$$

Obtain and interpret the expected value of the random variable X.

6. The time to failure in thousands of hours of an electronic equipment used in a manufactured computer has the density function

$$f(x) = \begin{cases} 3e^{-3x} & x > 0 \\ 0 & \text{elsewhere} \end{cases}.$$

Find the expected life of this electronic equipment.

7. The probability density function of the random variable X is given by

$$f(x) = \begin{cases} 16\,x\,e^{-4x} & \text{for } x > 0 \\ 0 & \text{for } x \le 0 \end{cases}$$

find the mean and variance of X.

8. A lottery with 600 tickets gives one prize of ₹200, four prizes of ₹100, and six prizes of ₹ 50. If the ticket costs is ₹ 2, find the expected profit amount of a ticket.

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